# Apache Airflow

Лекция 2

# Batch vs Online

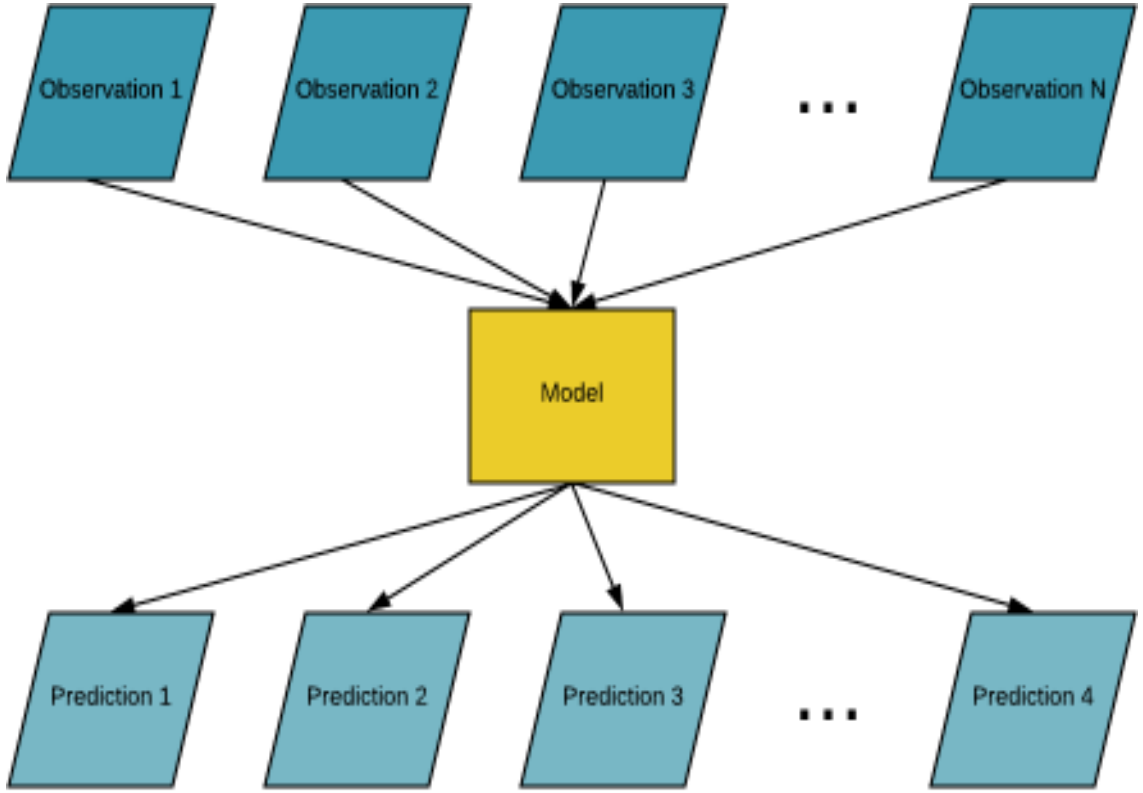



Figure 1. Batch Inference

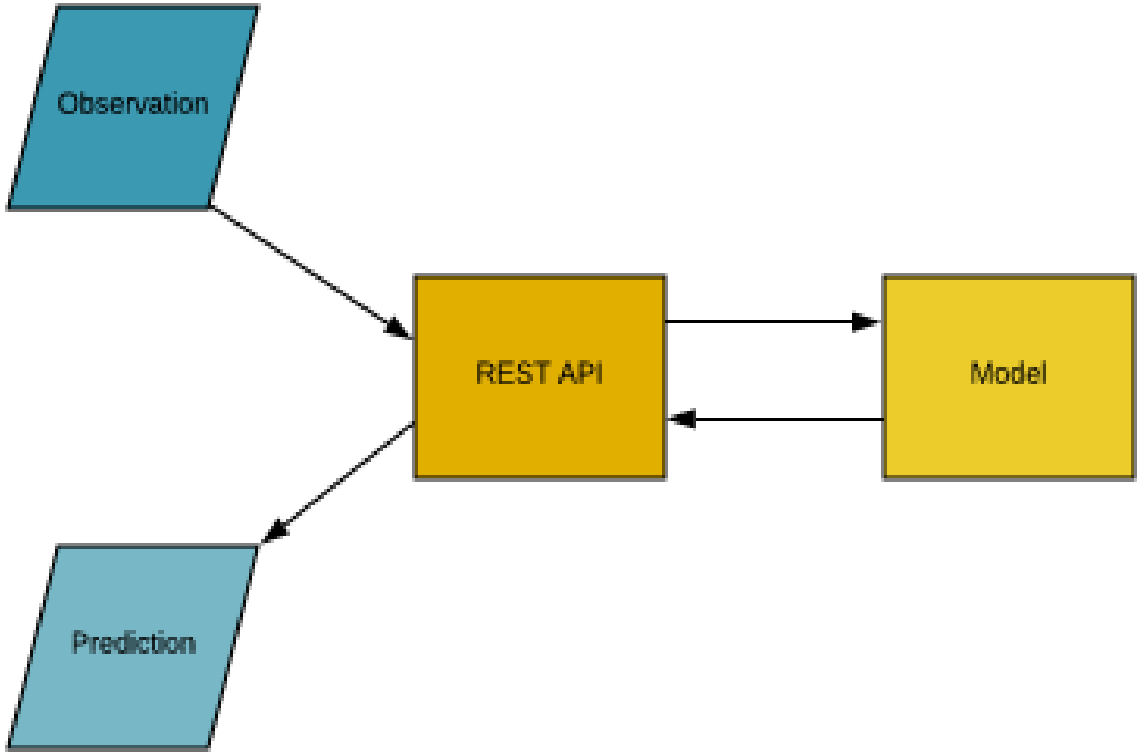



Figure 2. Online Inference

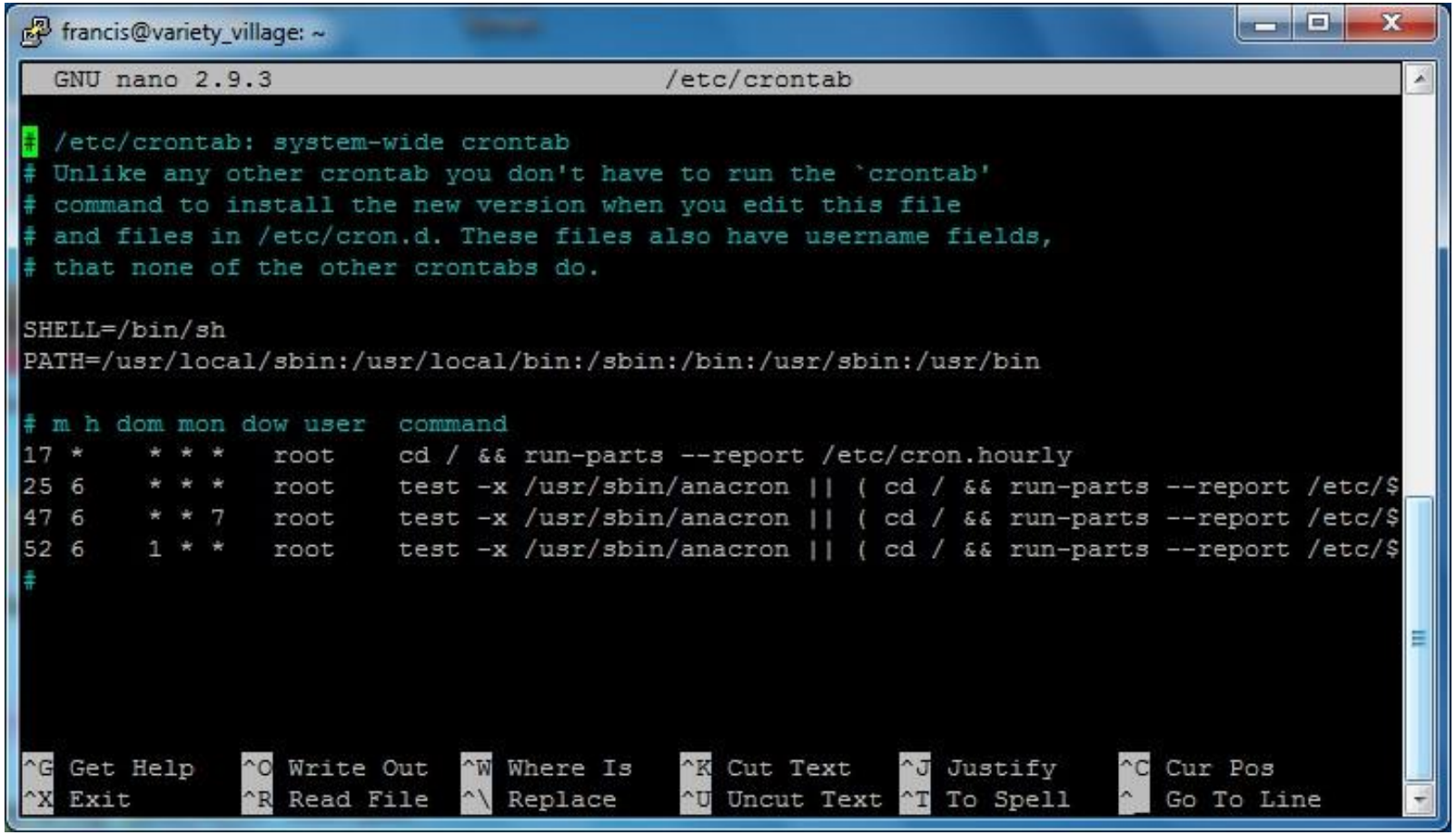
# Cron

```
# * * * * *  command to execute
# ┬ ┬ ┬ ┬ ┬
# │ │ │ │ │
# │ │ │ │ └──── day of week (0 - 7)
# │ │ │ └─────────── month (1 - 12)
# │ │ └──────────────── day of month (1 - 31)
# │ └──────────────────── hour (0 - 23)
# └───────────────────────── min (0 - 59)
```

CRON job

# Cron

https://crontab.guru/

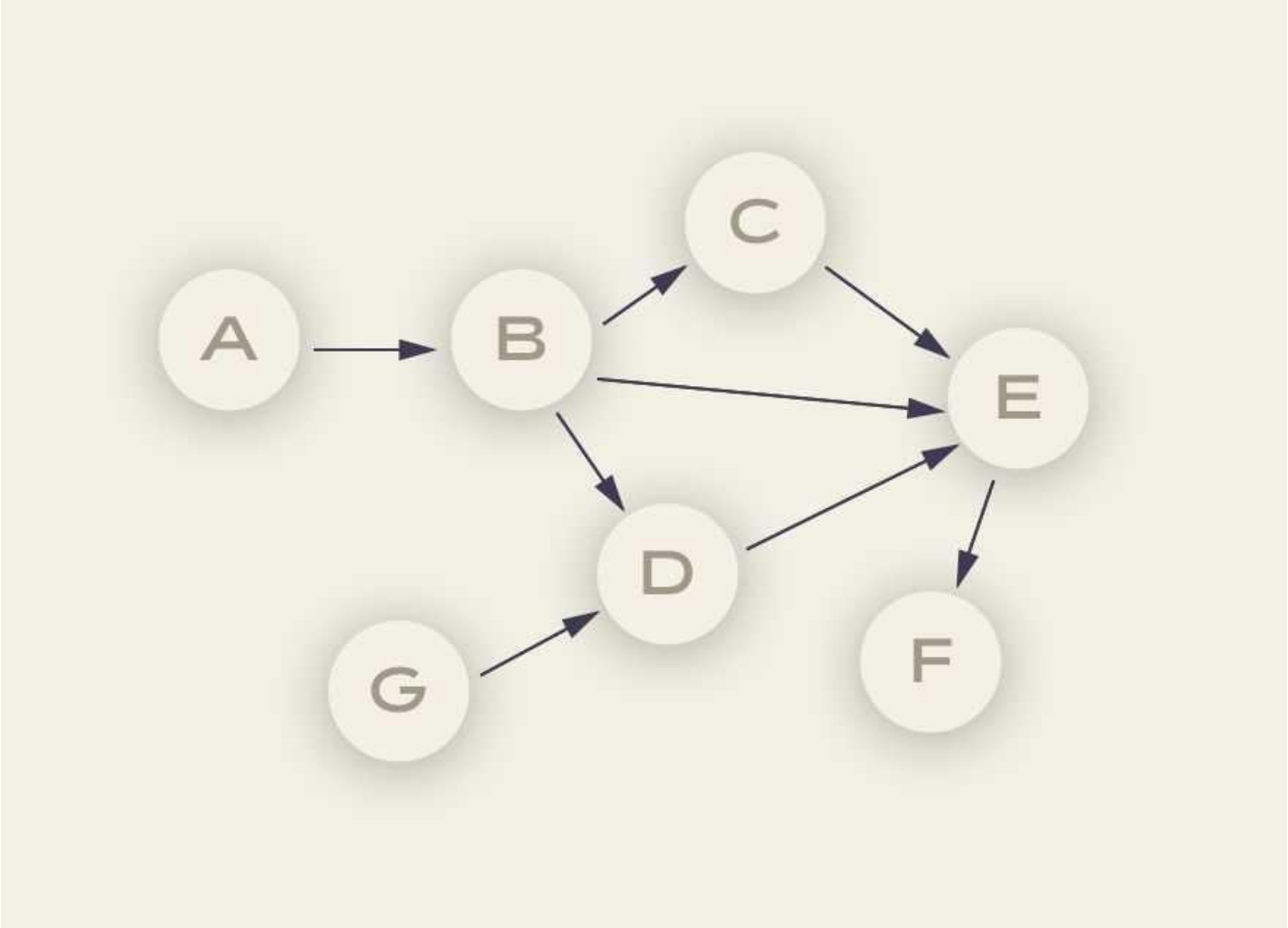


# Оркестратор данных

Фремворк, который умеет:

1. Планирование задач (когда запустить)
2. Управление зависимостями (что за чем запустить)
3. Репроцессинг (легко перезапускать упавшее)
4. Мониторинг (рассылка уведомлений в случае падения)

# DAG



# Apache Airflow

https://airflow.apache.org/

https://github.com/apache/airflow

# Создание DAG

```python
import datetime

from airflow import DAG
from airflow.operators.empty import EmptyOperator

with DAG(
    dag_id="my_dag_name",
    start_date=datetime.datetime(2021, 1, 1),
    schedule="@daily",
):
    EmptyOperator(task_id="task")
```

```python
import datetime

from airflow import DAG
from airflow.operators.empty import EmptyOperator

my_dag = DAG(
    dag_id="my_dag_name",
    start_date=datetime.datetime(2021, 1, 1),
    schedule="@daily",
)
EmptyOperator(task_id="task", dag=my_dag)
```

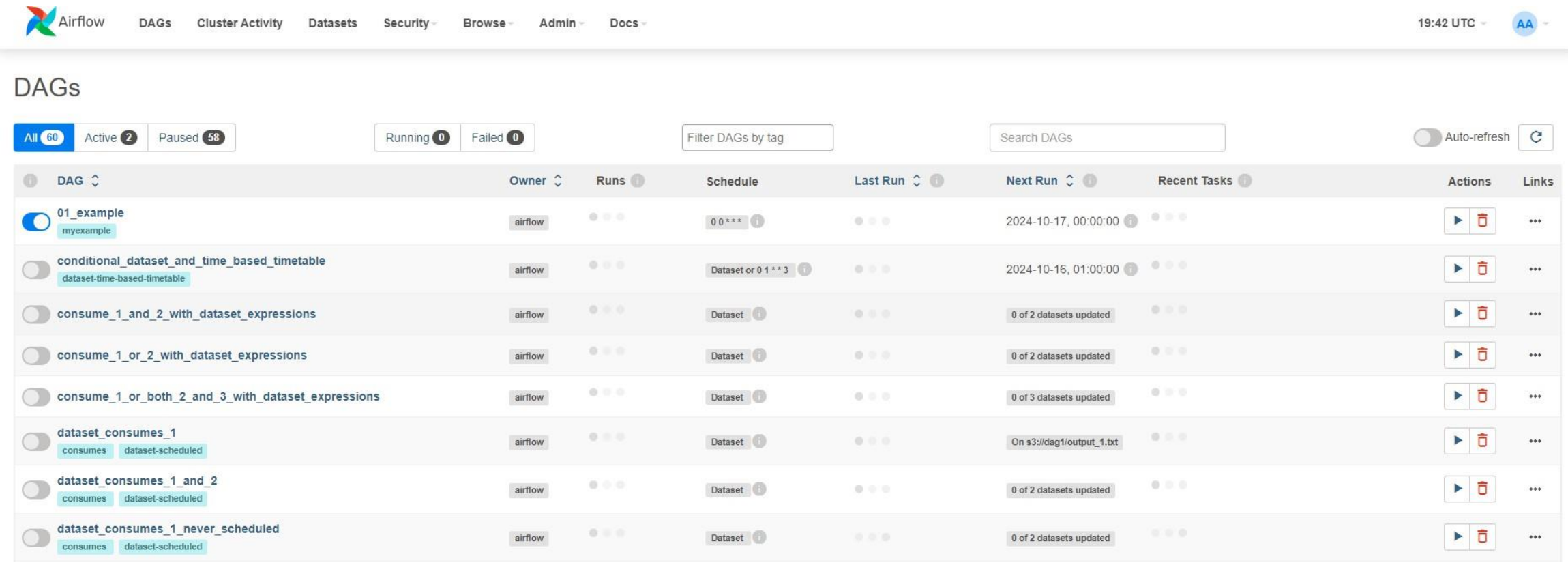
```python
import datetime

from airflow.decorators import dag
from airflow.operators.empty import EmptyOperator


@dag(start_date=datetime.datetime(2021, 1, 1), schedule="@daily")
def generate_dag():
    EmptyOperator(task_id="task")


generate_dag()
```

# Интерфейс DAG



# Airflow Task

Базовая единица DAG

Может быть в виде:

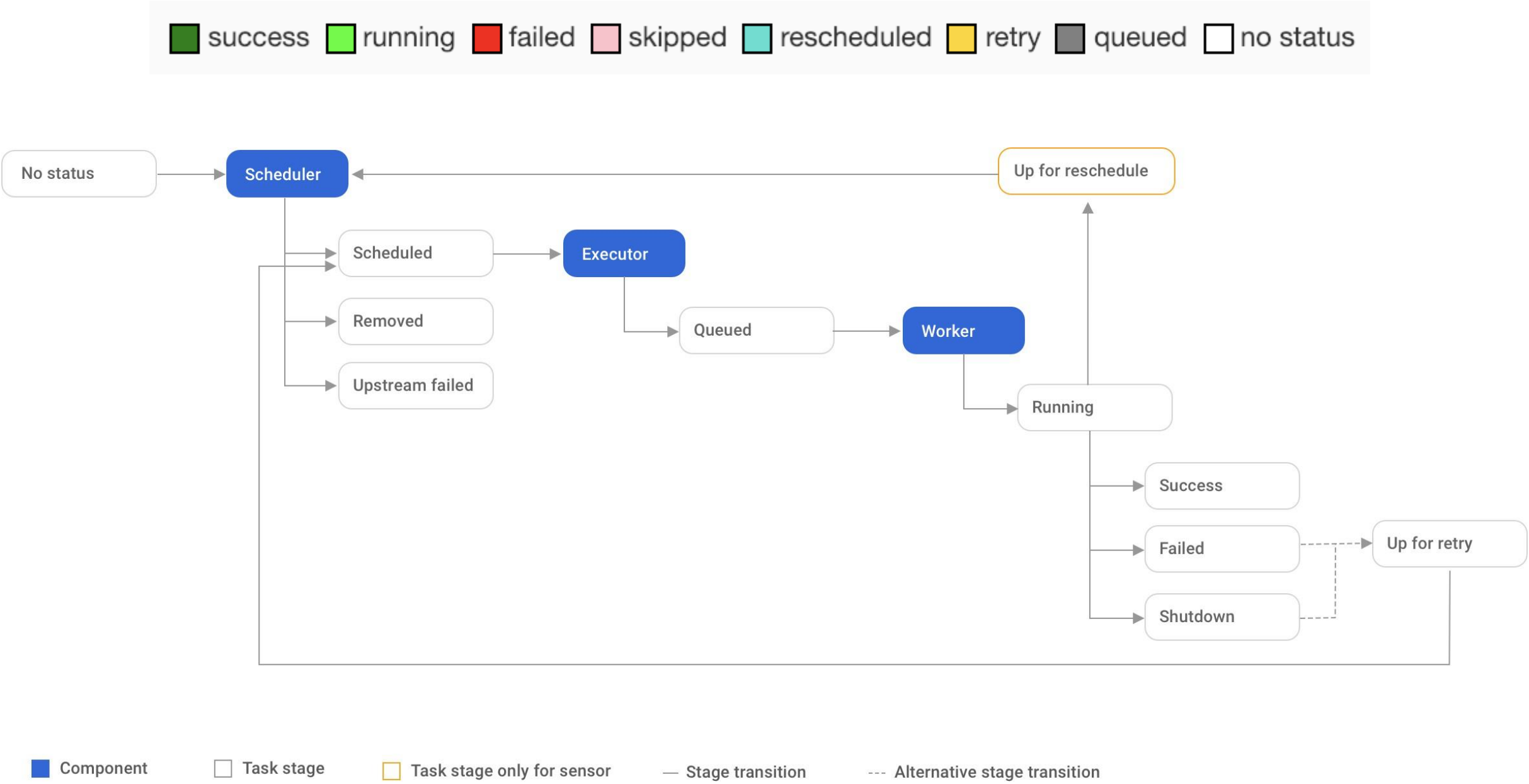
1. Оператор
2. Сенсор
3. TaskFlow декоратор @task

```
first_task >> second_task >> [third_task, fourth_task]
```

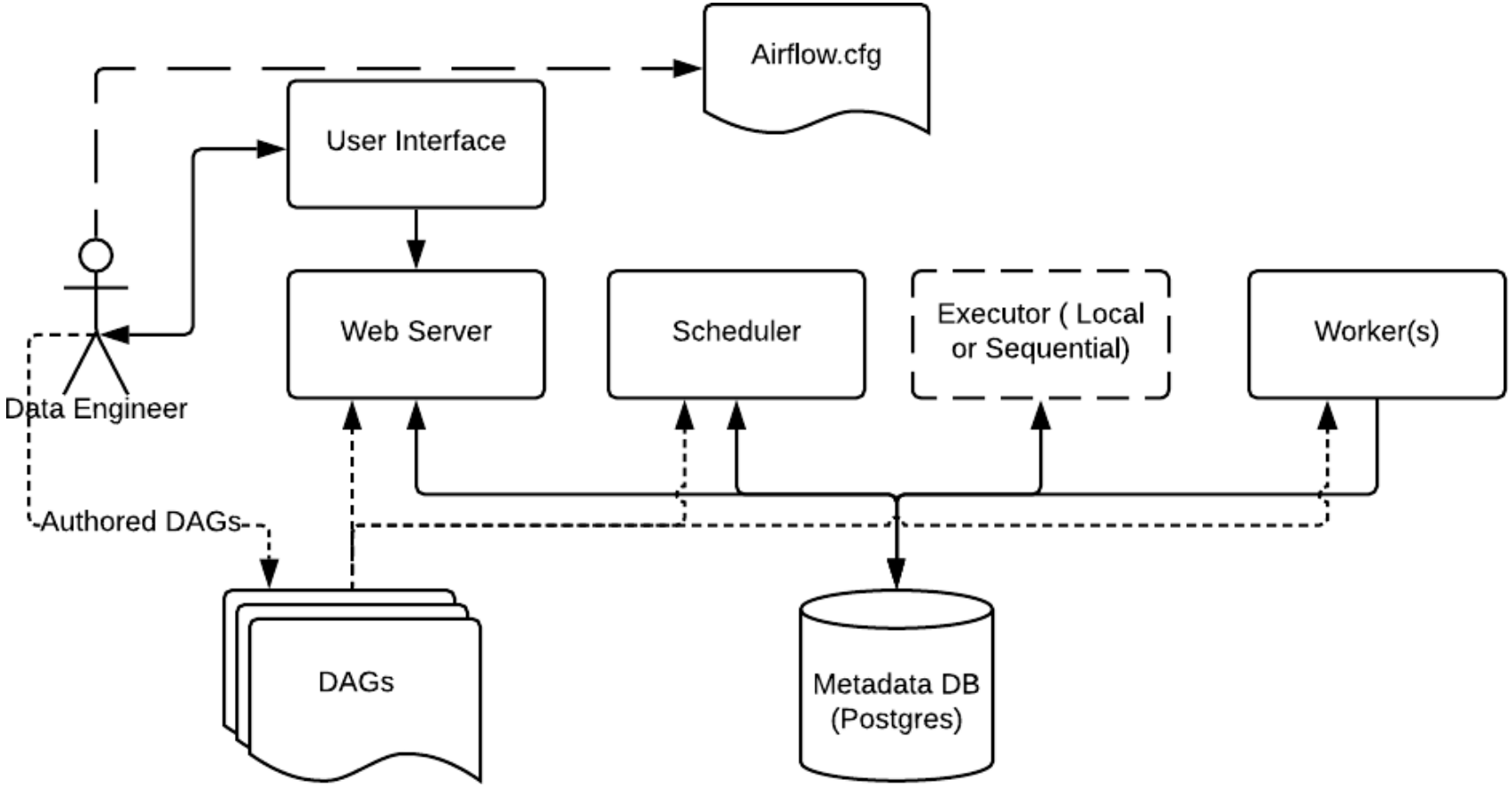
```
first_task.set_downstream(second_task)
third_task.set_upstream(second_task)
```

# Airflow Task состояния

- `none` : The Task has not yet been queued for execution (its dependencies are not yet met)
- `scheduled` : The scheduler has determined the Task's dependencies are met and it should run
- `queued` : The task has been assigned to an Executor and is awaiting a worker
- `running` : The task is running on a worker (or on a local/synchronous executor)
- `success` : The task finished running without errors
- `restarting` : The task was externally requested to restart when it was running
- `failed` : The task had an error during execution and failed to run
- `skipped` : The task was skipped due to branching, LatestOnly, or similar.
- `upstream_failed` : An upstream task failed and the Trigger Rule says we needed it
- `up_for_retry` : The task failed, but has retry attempts left and will be rescheduled.
- `up_for_reschedule` : The task is a Sensor that is in `reschedule` mode
- `deferred` : The task has been deferred to a trigger
- `removed` : The task has vanished from the DAG since the run started


success
running
failed
skipped
rescheduled
retry
queued
no status
No status
Scheduler
Up for reschedule
Scheduled
Executor
Removed
Queued
Worker
Upstream failed
Running
Success
Failed
Up for retry
Shutdown
Component
Task stage
Task stage only for sensor
Stage transition
Alternative stage transition

# Airflow Concepts



# Airflow Task Operator

```python
task_bash_op = BashOperator(
    task_id="show_files",
    bash_command="ls /opt/airflow/",
)

email = EmailOperator(
    task_id='send_email',
    to="email@gmail.com",
    subject="Update complete",
    html_content='Hello'
)
```

```python
def my_custom_function(input_value):
    print(f'Square of input value is: {input_value ** 2}')

task_python_op = PythonOperator(
    task_id="python_operator",
    python_callable=my_custom_function,
    op_kwargs={"input_value": 15},
)
```

# Airflow Task Operator

**Возможные операторы:**

- SparkSubmitOperator
- PostgresOperator
- HiveOperator
- DockerOperator
- SSHOperator
- TelegramOperator

```python
from airflow.models.baseoperator import BaseOperator


class HelloOperator(BaseOperator):
    def __init__(self, name: str, **kwargs) -> None:
        super().__init__(**kwargs)
        self.name = name

    def execute(self, context):
        message = f"Hello {self.name}"
        print(message)
        return message
```

# Providers

Active providers

- Airbyte
- Alibaba
- Amazon
- Apache Beam
- Apache Cassandra
- Apache Drill
- Apache Druid
- Apache Flink
- Apache HDFS
- Apache Hive
- Apache Iceberg
- Apache Impala
- Apache Kafka
- Apache Kylin
- Apache Livy
- Apache Pig
- Apache Pinot
- Apache Spark
- Apprise
- ArangoDB
- Asana
- Atlassian Jira
- Celery
- Cloudant
- CNCF Kubernetes
- Cohere
- Common Compat
- Common IO
- Common SQL
- Databricks
- Datadog
- dbt Cloud
- Dingding
- Discord
- Docker
- Elasticsearch
- Exasol
- FAB (Flask-AppBuilder)
- Facebook
- File Transfer Protocol (FTP)
- GitHub
- Google
- gRPC
- Hashicorp
- Hypertext Transfer Protocol (HTTP)
- IBM Cloudant
- Influx DB
- Internet Message Access Protocol (IMAP)
- Java Database Connectivity (JDBC)
- Jenkins
- Microsoft Azure
- Microsoft SQL Server (MSSQL)
- Microsoft PowerShell Remoting Protocol (PSRP)
- Microsoft Windows Remote Management (WinRM)
- MongoDB
- MySQL
- Neo4j
- ODBC
- OpenAI
- OpenFaaS
- OpenLineage
- Open Search
- Opsgenie
- Oracle
- Pagerduty
- Papermill
- PgVector
- Pinecone
- PostgreSQL
- Presto
- Qdrant
- Redis
- Salesforce
- Samba
- Segment
- Sendgrid
- SFTP
- Singularity
- Slack
- SMTP
- Snowflake
- SQLite
- SSH
- Tableau
- Telegram
- Teradata
- Trino
- Vertica
- Weaviate
- Yandex
- YDB
- Zendesk

# XCom

List XComs

Search

« < 1 2 > » Page size Actions

Record Count: 125

| | Key | Value | Timestamp | Dag Id | Task Id | Run Id | Map Index | Execution Date |
|---|---|---|---|---|---|---|---|---|
| | return_value | plugins | 2024-10-18, 09:04:03 | 01_example | show_files | scheduled__2024-10-17T00:00:00+00:00 | | 2024-10-17, 00:00:00 |
| | return_value | Whatever you return gets printed in the logs | 2024-10-18, 09:04:03 | 02_python_operator | print_the_context | scheduled__2024-10-17T00:00:00+00:00 | | 2024-10-17, 00:00:00 |
| | return_value | less_0.5 | 2024-10-18, 09:04:03 | 03_branching | branching | scheduled__2024-10-17T00:00:00+00:00 | | 2024-10-17, 00:00:00 |
| | return_value | Number is less than 0.5 | 2024-10-18, 09:04:09 | 03_branching | less_0.5 | scheduled__2024-10-17T00:00:00+00:00 | | 2024-10-17, 00:00:00 |
| | skipmixin_key | {'followed': ['less_0.5']} | 2024-10-18, 09:04:03 | 03_branching | branching | scheduled__2024-10-17T00:00:00+00:00 | | 2024-10-17, 00:00:00 |
| | return_value | Number is more than 0.5 | 2024-10-17, 21:00:28 | 03_branching | more_0.5 | scheduled__2024-10-14T00:00:00+00:00 | | 2024-10-14, 00:00:00 |
| | skipmixin_key | {'followed': ['more_0.5']} | 2024-10-17, 21:00:20 | 03_branching | branching | scheduled__2024-10-14T00:00:00+00:00 | | 2024-10-14, 00:00:00 |
| | return_value | more_0.5 | 2024-10-17, 21:00:20 | 03_branching | branching | scheduled__2024-10-14T00:00:00+00:00 | | 2024-10-14, 00:00:00 |
| | return_value | less_0.5 | 2024-10-17, 21:00:21 | 03_branching | branching | scheduled__2024-10-13T00:00:00+00:00 | | 2024-10-13, 00:00:00 |
| | return_value | Number is less than 0.5 | 2024-10-17, 21:00:27 | 03_branching | less_0.5 | scheduled__2024-10-13T00:00:00+00:00 | | 2024-10-13, 00:00:00 |
| | skipmixin_key | {'followed': ['less_0.5']} | 2024-10-17, 21:00:21 | 03_branching | branching | scheduled__2024-10-13T00:00:00+00:00 | | 2024-10-13, 00:00:00 |
| | skipmixin_key | {'followed': ['less_0.5']} | 2024-10-17, 21:00:20 | 03_branching | branching | scheduled__2024-10-12T00:00:00+00:00 | | 2024-10-12, 00:00:00 |
| | return_value | Number is less than 0.5 | 2024-10-17, 21:00:27 | 03_branching | less_0.5 | scheduled__2024-10-12T00:00:00+00:00 | | 2024-10-12, 00:00:00 |
| | return_value | less_0.5 | 2024-10-17, 21:00:20 | 03_branching | branching | scheduled__2024-10-12T00:00:00+00:00 | | 2024-10-12, 00:00:00 |

# Templating

| Variable | Type | Description |
|---|---|---|
| `{{ data_interval_start }}` | pendulum.DateTime | Start of the data interval. Added in version 2.2. |
| `{{ data_interval_end }}` | pendulum.DateTime | End of the data interval. Added in version 2.2. |
| `{{ logical_date }}` | pendulum.DateTime | A date-time that logically identifies the current DAG run. This value does Use `data_interval_start` and `data_interval_end` instead if you want such as to get a slice of rows from the database based on timestamps. |
| `{{ ds }}` | str | The DAG run's logical date as `YYYY-MM-DD`.<br>Same as `{{ logical_date \| ds }}`. |
| `{{ ds_nodash }}` | str | Same as `{{ logical_date \| ds_nodash }}`. |
| `{{ exception }}` | None \| str \| Exception KeyboardInterrupt | Error occurred while running task instance. |
| `{{ ts }}` | str | Same as `{{ logical_date \| ts }}`.<br>Example: `2018-01-01T00:00:00+00:00`. |
| `{{ ts_nodash_with_tz }}` | str | Same as `{{ logical_date \| ts_nodash_with_tz }}`.<br>Example: `20180101T000000+0000`. |
| `{{ ts_nodash }}` | str | Same as `{{ logical_date \| ts_nodash }}`.<br>Example: `20180101T000000`. |
| `{{ prev_data_interval_start_success }}` | pendulum.DateTime \| `None` | Start of the data interval of the prior successful `DagRun`.<br>Added in version 2.2. |
| `{{ prev_data_interval_end_success }}` | pendulum.DateTime \| `None` | End of the data interval of the prior successful `DagRun`.<br>Added in version 2.2. |
| `{{ prev_start_date_success }}` | pendulum.DateTime \| `None` | Start date from prior successful `DagRun` (if available). |
| `{{ prev_end_date_success }}` | pendulum.DateTime \| `None` | End date from prior successful `DagRun` (if available). |
| `{{ inlets }}` | list | List of inlets declared on the task. |

# Sensors

Примеры:

- Появление файла
- Появление файла на HDFS
- Код запроса 200 от http запроса к сервису
- Выполнение SQL-запроса
- Создание таблицы

Режимы:

1. Poke
2. Reschedule

## airflow.sensors

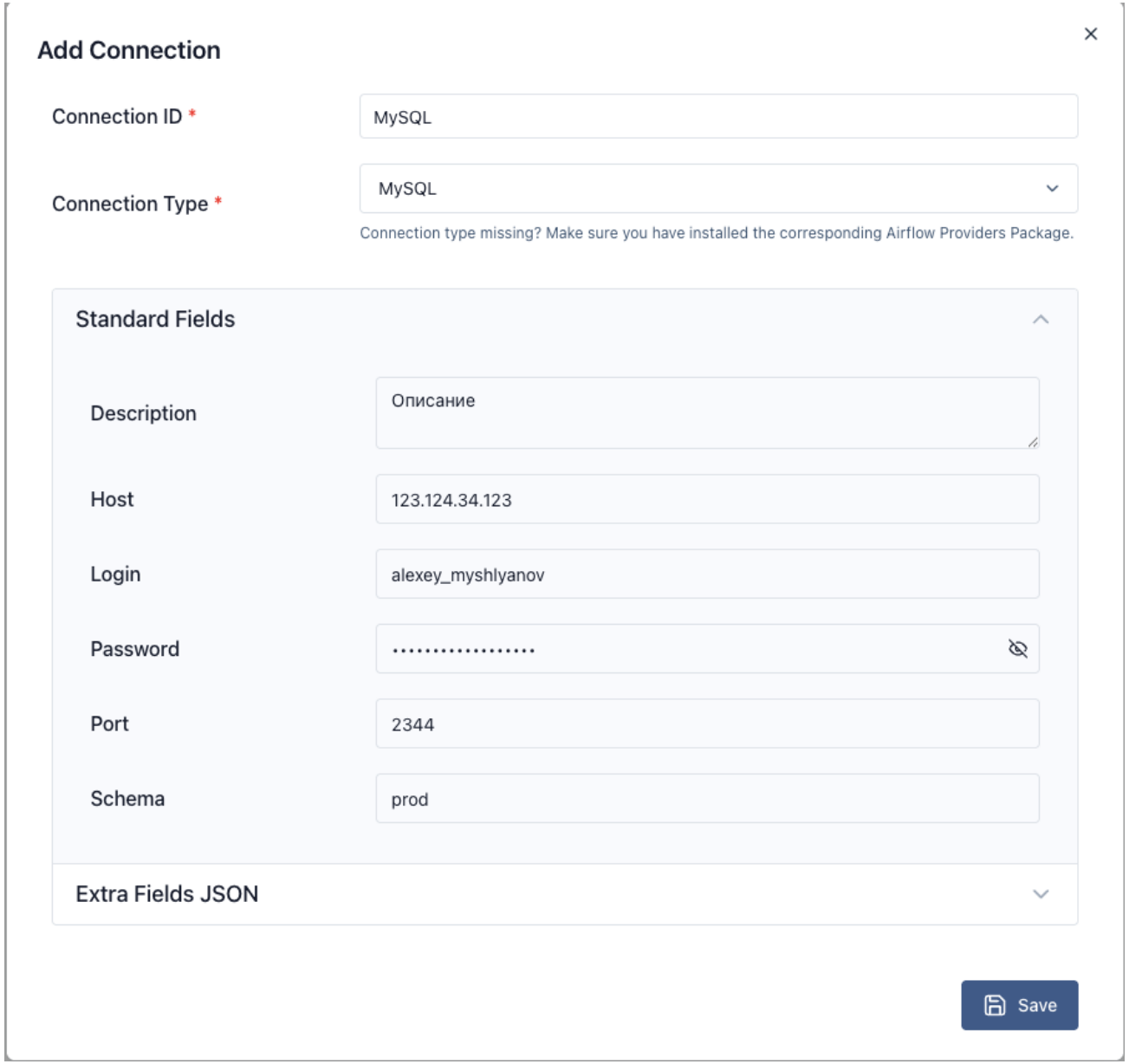
Sensors.

```
sphinx-autoapi-skip
```

### Submodules

- airflow.sensors.base
- airflow.sensors.bash
- airflow.sensors.date_time
- airflow.sensors.external_task
- airflow.sensors.filesystem
- airflow.sensors.python
- airflow.sensors.time_delta
- airflow.sensors.time_sensor
- airflow.sensors.weekday

# Connections



# Домашнее задание №1

Soft deadline – 20.03.2026 **(+5 баллов)**

Hard deadline – 05.06.2026

Нужно реализовать ETL процесс. Функциональные требования:

1. Инкрементальная загрузка. Необходимо загружать данные только за определенный день. Если файла за дату нет, то DAG завершается успешно, ничего не сделав. При повторном запуске должен вести себя корректно (как именно корректно – решать вам) **(2 балла)**

# Домашнее задание №1

2. Два слоя хранения. Сырые данных хранятся в папке raw, обработанные в processed **(2 балла)**

3. Идемпотентность. Повторные запуски/перезапуски или запуски задним числом должны работать корректно **(2 балла)**

4. Проверка качества данных:

   а) Не пустая таблица на выходе

   б) Нет строк с transaction_id IS NULL

   в) Нет строк с отрицательным значением amount

   г) transaction_id является уникальным
   **(2 балла)**

# Домашнее задание №1

5) Учесть, что DAG может падать. Если падает, то перезапускать. Если умирает, то шлет письмо на почту. Разобраться с max_active_runs и concurrency **(1 балл)**

6) Очистка устаревших данных. Если данным (сырым) больше 2 недель, их надо удалить **(1 балл)**

**Итого: 10 баллов (15 в soft deadline)**